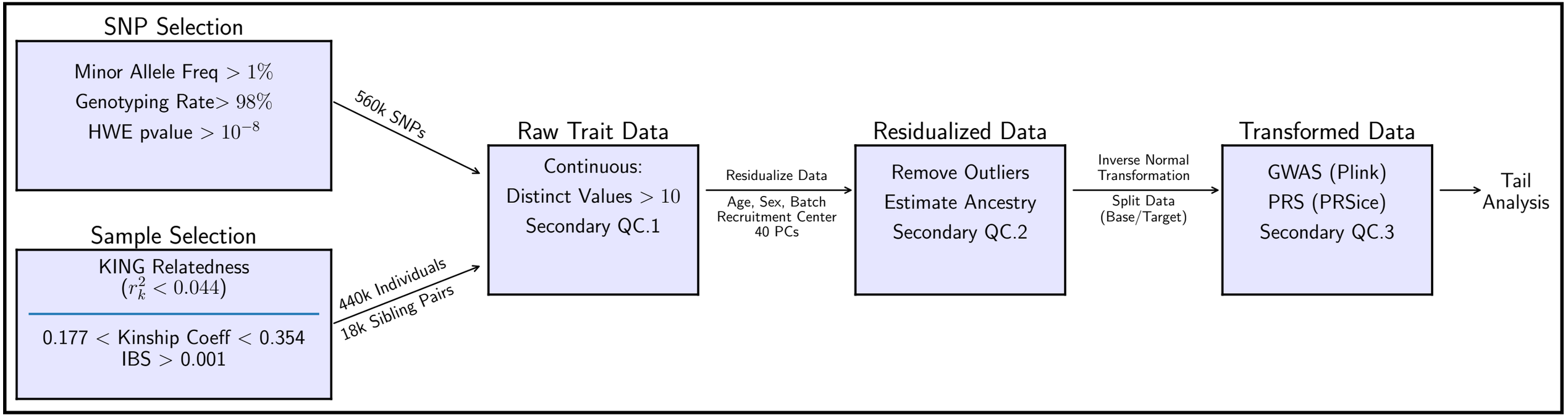
SNP Selection
Minor Allele Freq > 1%
Genotyping Rate> 98%
HWE pvalue > 10^{-8}
560k SNPs
Sample Selection
KING Relatedness
($r_k^2 < 0.044$)
0.177 < Kinship Coeff < 0.354
IBS > 0.001
440k Individuals
18k Sibling Pairs
Raw Trait Data
Continuous:
Distinct Values > 10
Secondary QC.1
Residualize Data
Age, Sex, Batch
Recruitment Center
40 PCs
Residualized Data
Remove Outliers
Estimate Ancestry
Secondary QC.2
Inverse Normal
Transformation
Split Data
(Base/Target)
Transformed Data
GWAS (Plink)
PRS (PRSice)
Secondary QC.3
Tail
Analysis